# APS110H1f2026 - September 24, 2026

September 24, 2026 8:05 AM

- Properties of Matter(s) - today again (and also for this week upcoming) - Structure-Property Relationship
  - Calculation
  - Nano structure

Mass and Mass to atoms

September 24, 2026 8:12 AM

u = Dalton – periodic table

(1) $u = 1\,kg \times 6.02214 \times 10^{26}$

$1\,kg = u \times 1.66054 \times 10^{-27}$

(2) $\frac{\text{mass of } u}{\text{mass of atom}} = \#\text{ of atoms.}$ only a metal

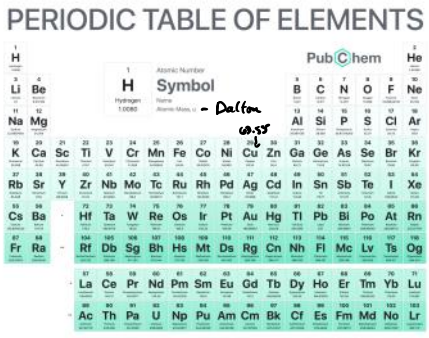
atom = 1u

(3) $\frac{\text{mass of } u}{\text{mass of atom(a)} + \text{mass of atom(b)}} = \#\text{ of atoms}$

ceramics

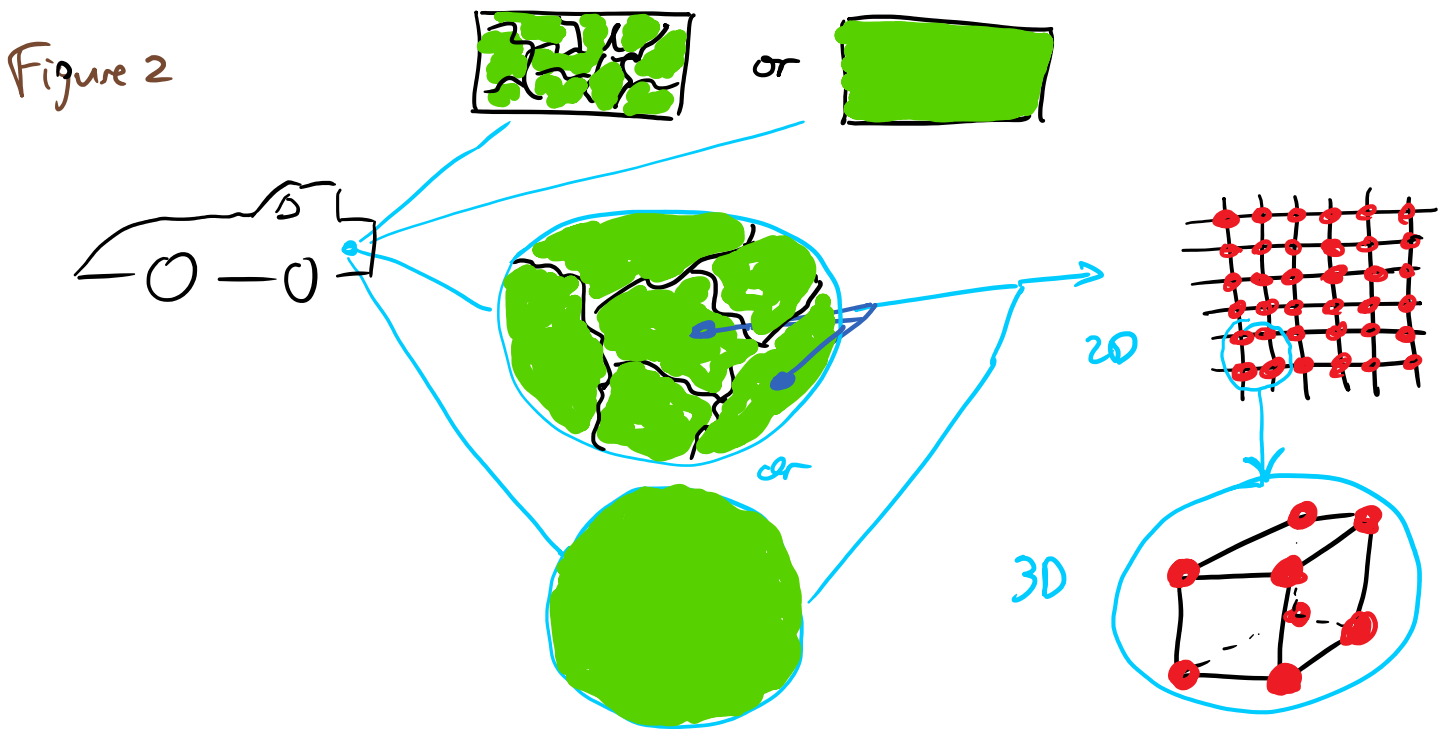
polymers + plastic

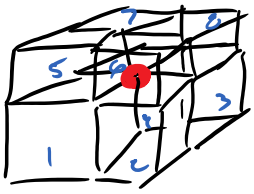
(4) $\frac{\text{mass of } u}{\text{mass of atom(a)} + \text{mass of atom(b)} + \text{mass of atom(c)}} = \#\text{ of atoms}$

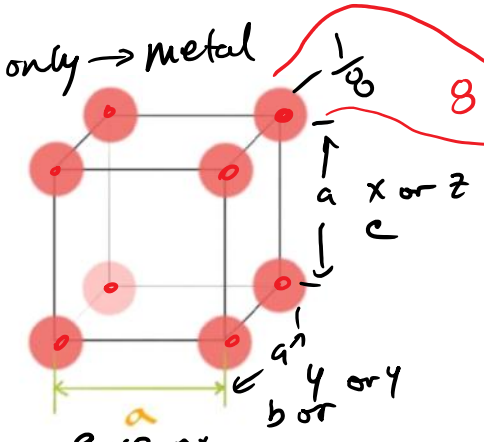


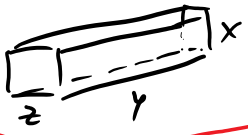
# Manufacturing - 2D + 3D structure - onto Nano Structure

September 24, 2026 8:33 AM








example only...

8 × 1/8 number of atoms = 1

$$m = n \times \frac{A}{N_A}$$

← molar mass of atom or can be U ↑ Daltons

↑ mass ↑ number of atoms

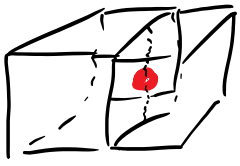
**Figure 4:** The simple cubic unit cell, showing an atom centred at each of the eight corners of a cube. The lighter red dots are the atoms located on the hidden parts of the cube. The cube edge length is known as the lattice parameter and is denoted with the letter a.

Figure 4

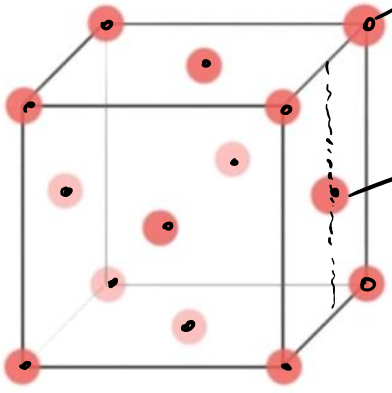
$N_A$ - Avagadro's

$6.023 \times 10^{23}$

$6.02214076 \times 10^{23}$



metal.



1/8 x 8. = 1

1/2 x 6 = 3

⇒ "4 atoms"

(a) $m = n \times \frac{A}{N_A}$

(b) density – tomorrow.

**Figure 5:** The FCC unit cell, showing an atom centred at each of the eight corners of a cube as well as one atom centred at the very centre of each of the six cube faces. The lighter red dots are the atoms located on the hidden parts of the cube.

Figure 5